IOT PLATFORM

Introduction

Goal: Develop an IoT-based platform to support scalable, continuous releases with high availability.

Summary:

- Enables subsequent releases and IoT scenario extensions.
- Built on DevOps principles with automated CI/CD pipeline.
- Split across cloud and on-prem environments between two AKS clusters.

SYSTEM ARCHITECTURE

IoT Devices (publishers): reads sensor data and publishes to an MQTT topic.

MQTT Broker: receives messages and forwards them to Kafka, adding room identifiers and validating data fields.

Kafka: streams data for processing, Kafka consumer reads data from Kafka and exposes it as metrics to Prometheus

Prometheus and Grafana: monitors metrics and visualises the data

CI/CD Pipeline Overview

Pipeline Stages:

- **Build & Push:** Automatically builds the images, pushes them to DockerHub and updates the pipeline.
- **Testing:** ArgoCD deploys the Docker images from Docker Hub to a test cluster. The pipeline comes into the cluster, run the system tests. Once the tests pass, ArgoCD updates the manifests and promotes the deployment to a production cluster.
- **Production Deployment:** ArgoCD manages releases, promoting changes across dev, test, and deployment.

System Testing and Validation

SystemTesting for Data Flow

Publishers: Send test messages to MQTT broker.

Data Validation: Messages flow through Kafka to Prometheus, validating that data appears as expected.

Prometheus Check: System test confirms data visibility in Prometheus.

Environment Structure

Environment Breakdown:

Dev:

- Hosts all active development code in Docker containers.
- Simulates the entire data pipeline locally, from data publishers through to Prometheus.

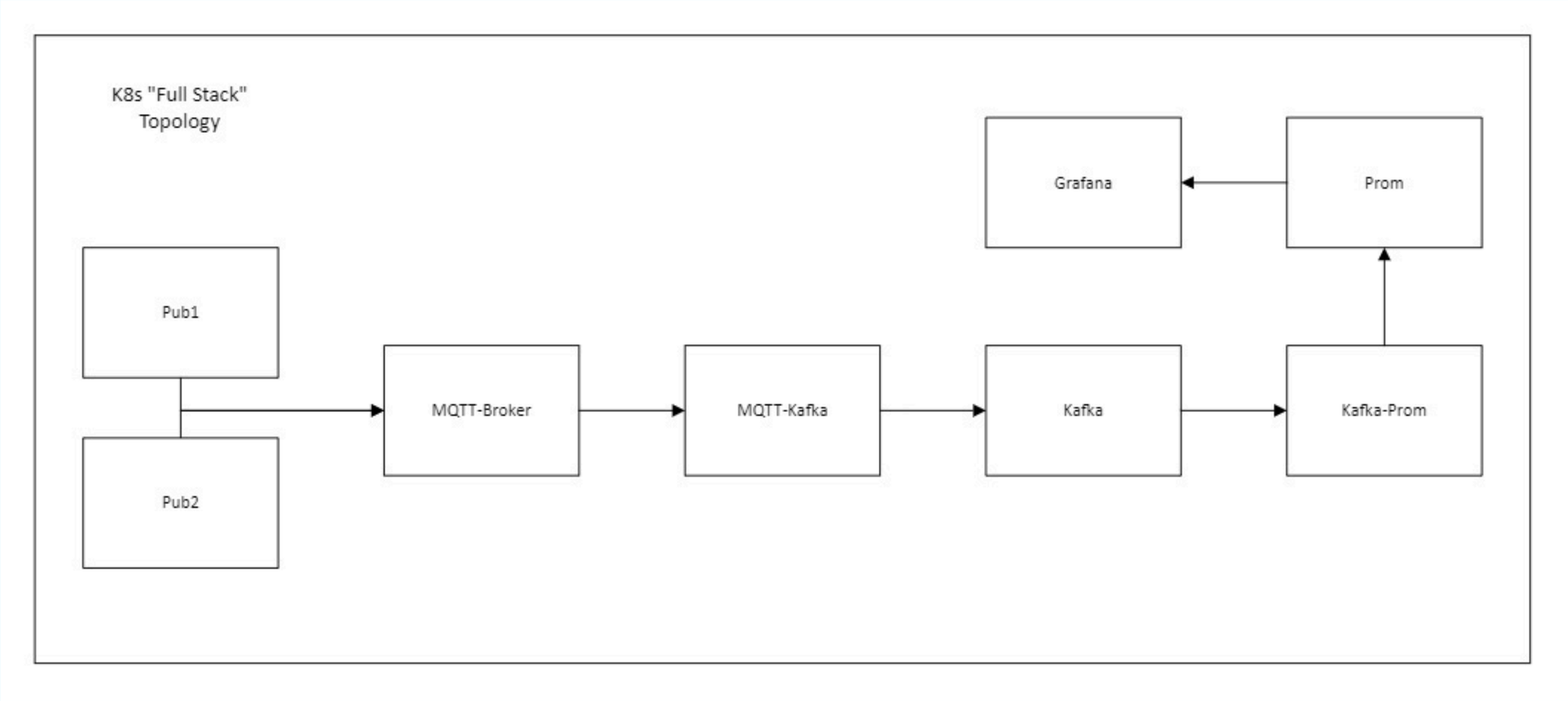
Depl:

- Split into on-prem and cloud deployments.
- On-Prem: Hosts components up to the MQTT broker.
- Cloud: Manages everything from Kafka onwards, completing the data flow.

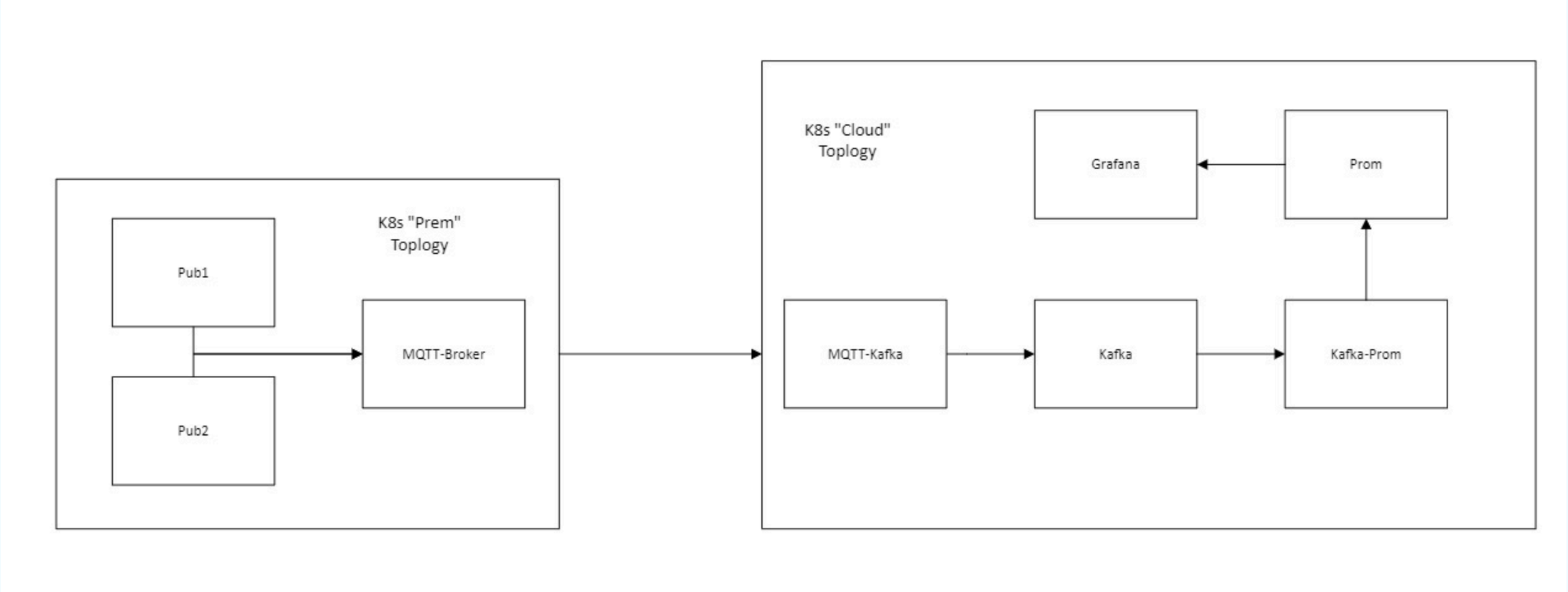
Test:

- Used exclusively for automated system testing.
- Ensures end-to-end functionality across the full pipeline.

K8s Full Stack Topology



Prem and Cloud Topology



GitOps and ArgoCD Integration

GitOps Principles: Git as the single source of truth for development and deployment of platform.

ArgoCD Workflow: Automatically syncs changes from Git to AKS clusters, managing deployments.

Benefits: Ensures version control, rollback capabilities, and streamlined environment updates.

THANK YOU